COMP 3311
DATABASE MANAGEMENT SYSTEMS

LECTURE 3 EXERCISES
DATABASE DESIGN:
ENTITY-RELATIONSHIP (E-R) DATA MODEL

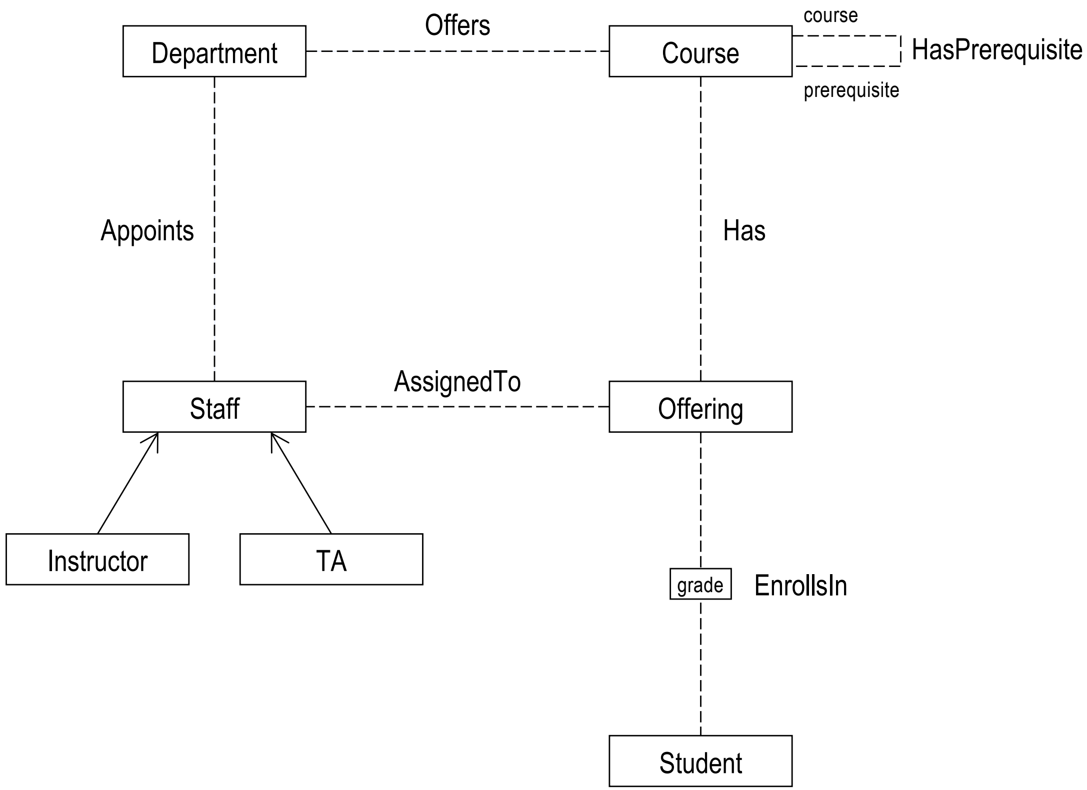
EXERCISE 1: UNIVERSITY APPLICATION

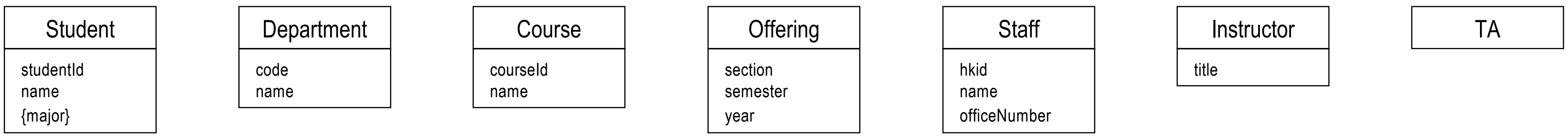
We want to record information about students, departments, courses and course teaching teams for a university.

- For each student we store the student id, name and majors.
- For each department we store a unique code and name.
- For each course we store a unique course id, name, department and prerequisites.
- For each offering of a course, we store the section, semester and year.
- Each student must enroll in one to five course offerings.
- Each course offering can enroll zero to sixty students.
- For each course offering that a student takes we store the grade.
- Each course offering's teaching team has one or more staff, who is either an instructor or a TA.
- For each staff assigned to a course offering's teaching team we store the hkid, name, department and office number.
- For each instructor we store their academic title (e.g., professor).

For the university application ERD, identify keys and discriminators of entities, identifying relationships of weak entities and show relationship cardinality and participation constraints.

EXERCISE 1: UNIVERSITY APPLICATION— E-R DIAGRAM





EXERCISE 1: UNIVERSITY APPLICATION— KEYS OF ENTITY TYPES

We want to record information about students, departments, courses and course teaching teams for a university.

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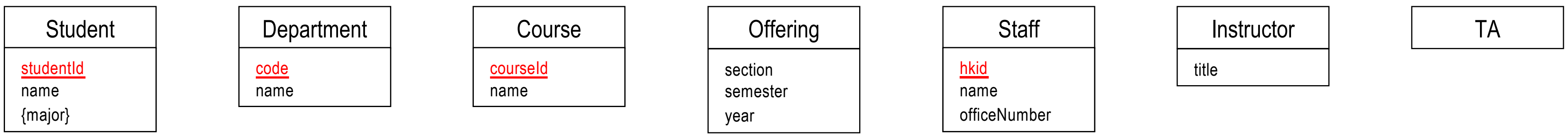
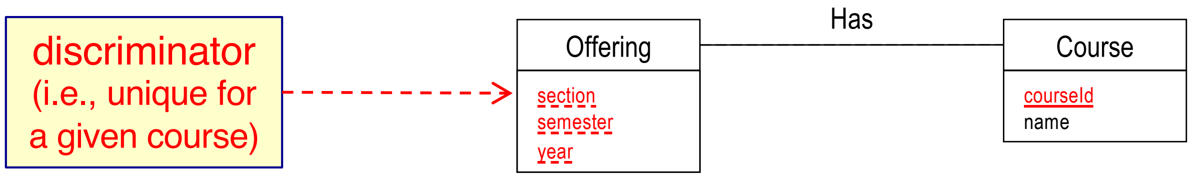


EXERCISE 1: UNIVERSITY APPLICATION— KEYS OF ENTITY TYPES

- For each offering of a course, we store the section, semester and year.



What kind of entity is **Offering**?

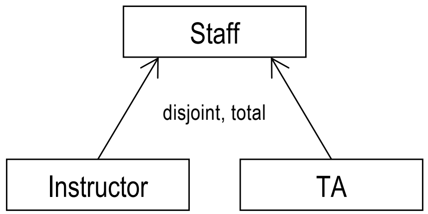
⟹ Weak entity dependent on **Course**

Is there a discriminator for **Offering**?

⟹ Yes — section, semester, year.

EXERCISE 1: UNIVERSITY APPLICATION— ENTITY GENERALIZATION COVERAGE

- Each course offering's teaching team has one or more **staff**, who is either an **instructor** or a **TA**.



What should be the disjointness constraint?

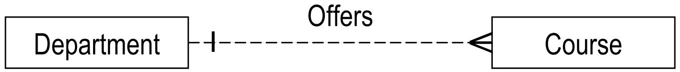
$\Longrightarrow$ disjoint

What should be the completeness constraint?

$\Longrightarrow$ total

EXERCISE 1: UNIVERSITY APPLICATION—RELATIONSHIP CARDINALITY & PARTICIPATION

- For each course we store a unique course id, name, department and prerequisites.



What should be the cardinality constraint (max-card) for **Department**?
⟹ many (A department can offer many courses—domain knowledge.)

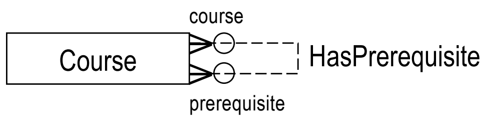
What should be the participation constraint (min-card) for **Department**?
⟹ unknown (Could be partial or total; need to verify with client. Leave unspecified.)

What should be the cardinality constraint (max-card) for **Course**?
⟹ unknown (Could be 1 or N; need to verify with client. Leave unspecified.)

What should be the participation constraint (min-card) for **Course**?
⟹ total (Every course must be offered by some department—domain knowledge.)

EXERCISE 1: UNIVERSITY APPLICATION— RELATIONSHIP CARDINALITY & PARTICIPATION

- For each course we store a unique course id, name, department and prerequisites.



What should be the cardinality constraints?

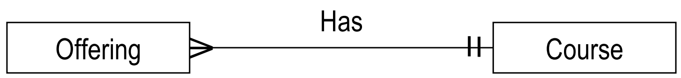
⟹ **Course** (prerequisite) many (A course can be a prerequisite for several courses.)
Course (course) many (A course can have several prerequisites.)

What should be the participation constraints?

⟹ **Course** (prerequisite) partial (A course does not have to be a prerequisite.)
Course (course) partial (A course can have no prerequisites.)

EXERCISE 1: UNIVERSITY APPLICATION— RELATIONSHIP CARDINALITY & PARTICIPATION

- For each offering of a course, we store the section, semester and year.



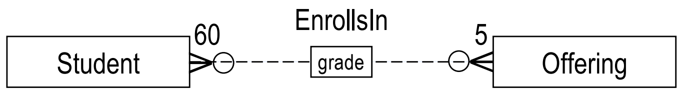
What should be the cardinality constraint (max-card) for **Offering**?
  ⟹ 1 (Every offering is for at most one course—domain knowledge.)

What should be the participation constraint (min-card) for **Offering**?
  ⟹ total (Every offering must be for some course—domain knowledge.)

What about for **Course**?
  ⟹ (?,many) (min-card most likely 0 but need to verify with client. Leave unspecified.)

EXERCISE 1: UNIVERSITY APPLICATION— RELATIONSHIP CARDINALITY & PARTICIPATION

- Each student must enroll in one to five course offerings.
- Each course offering can enroll zero to sixty students.



In the application domain, is a student required to be related to (enroll in) a course offering as soon as he/she becomes a student at the university?

No!
(domain knowledge)

Is **Offering** dependent on **Student**?
⟹ No.

What should be the cardinality constraint (max-card) for **Student**?
⟹ 5 A student can enroll in at most 5 course offerings.)

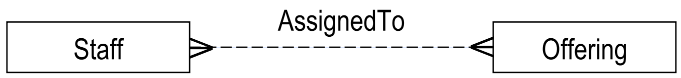
What should be the participation constraint (min-card) for **Student**?
⟹ total (A student must enroll in at least 1 course offering.)

What about for **Offering**?
⟹ (0, 60)

☞ **Does the participation constraint for Student make sense?**

EXERCISE 1: UNIVERSITY APPLICATION— RELATIONSHIP CARDINALITY & PARTICIPATION

- Each course offering's teaching team has one or more staff, who is either an instructor or a TA



In the application domain, is a course offering required to be related to (have assigned) a staff?

Need to verify with client!

Is **Offering** dependent on **Staff**?
⟹ No.

What should be the cardinality constraint (max-card) for **Offering**?
⟹ many (An offering can have several staff assigned to it.)

What should be the participation constraint (min-card) for **Offering**?
⟹ total (An offering has at least one staff assigned to it.)

What about for **Staff**?
⟹ (?,many) (min-card most likely 0 but need to verify with client. Leave unspecified.)

☞ **Does the participation constraint for Offering make sense?**

EXERCISE 1: UNIVERSITY APPLICATION— RELATIONSHIP CARDINALITY & PARTICIPATION

- For each staff assigned to a course offering's teaching team we store the hkid, name, department and office number.

What should be the cardinality constraint (max-card) for **Staff**?

⟹ 1 (For each staff … we store the … department ….)

What should be the participation constraint (min-card) for **Staff**?

⟹ total (Every staff must be appointed in some department—domain knowledge.)

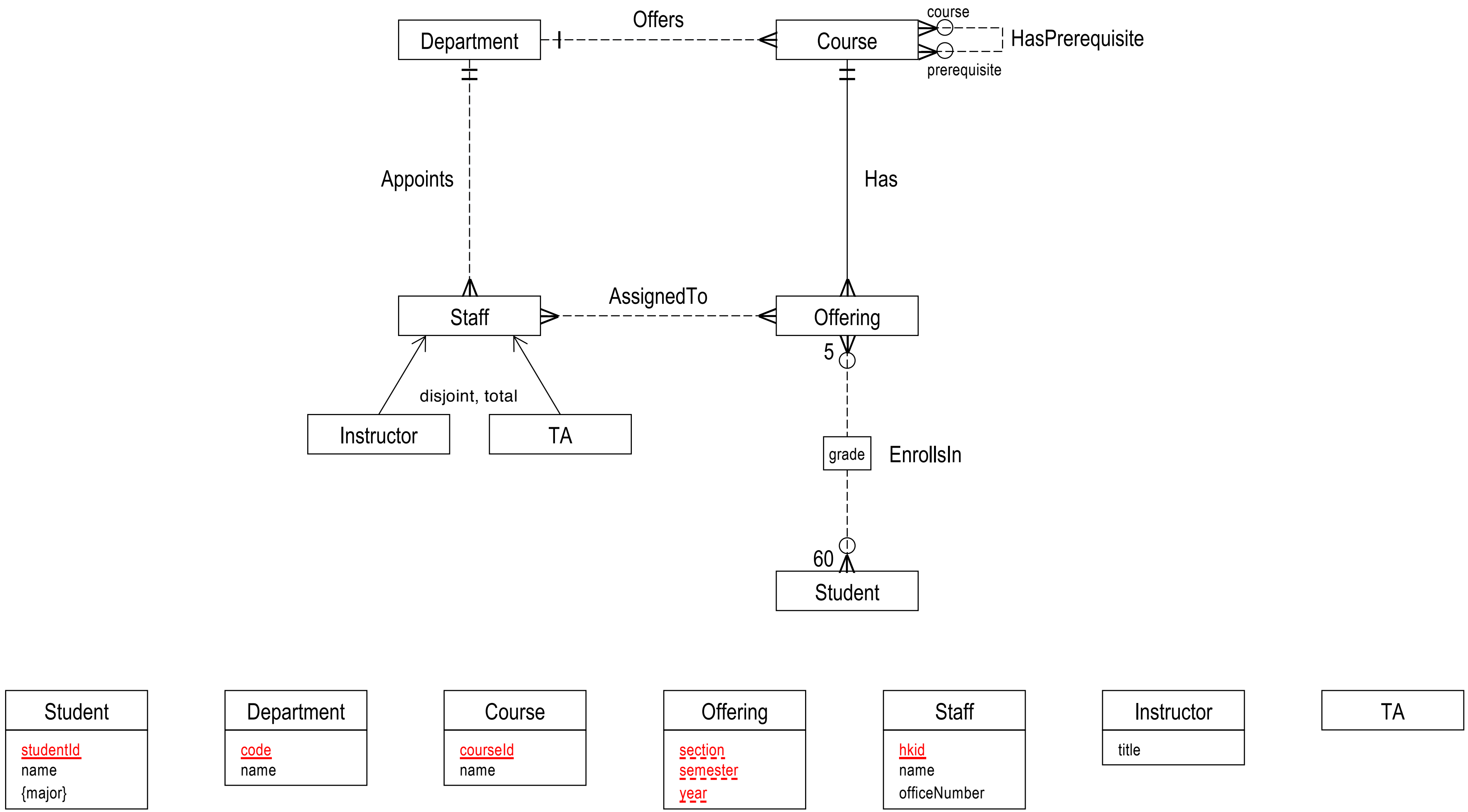
What should be the cardinality constraint (max-card) for **Department**?

⟹ many (A department can appoint several staff—domain knowledge.)

What should be the participation constraint (min-card) for **Department**?

⟹ unknown (Could be partial or total; need to verify with client. Leave unspecified.)

EXERCISE 1: UNIVERSITY APPLICATION—E-R DIAGRAM



EXERCISE 2: FACTORY APPLICATION

We want to record information about products that a factory manufactures.

- The factory has several employees. For each employee we store the employee id, name and salary.
- Each employee must be an admin staff or a worker, but not both.
- From time to time, admin staff are required to take seminars. For each seminar we store its id, name and date. For the admin staff, we store the grade received for each seminar taken.
- The factory manufactures several products; each product is identified by a product id and has a name.
- A worker is assigned to work on exactly one product; a product has multiple (one or more) workers assigned to it.
- Several parts are manufactured for each product. Each part has a part number and a color. Different parts of the same product have different part numbers. However, two parts that belong to different products may have the same part number.

For the factory application ERD, identify keys and discriminators of entities, identifying relationships of weak entities and show relationship cardinality and participation constraints.

EXERCISE 2: FACTORY APPLICATION— KEYS OF ENTITY TYPES

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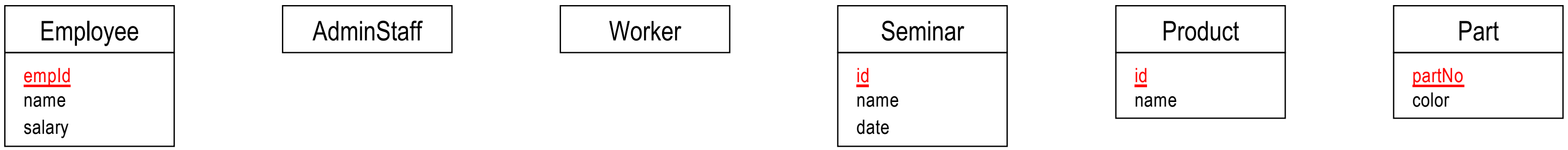
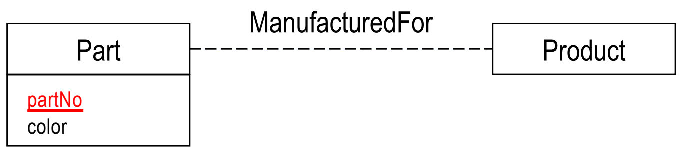
- Several parts are manufactured for each product. Each part has a part number and a color. Different parts of the same product have different part numbers. However, two parts that belong to different products may have the same part number.



EXERCISE 2: FACTORY APPLICATION—KEYS OF ENTITY TYPES

- Several parts are manufactured for each product. Each part has a part number and a color. Different parts of the same product have different part numbers. However, two parts that belong to different products may have the same part number.

  Is partNo a primary key?

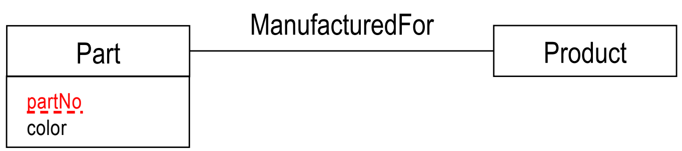


X

Consider: two parts that belong to different products may have the same part number.

⟹ **Part** is a weak entity dependent on Product.

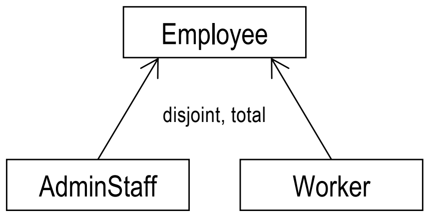
⟹ partNo is a discriminator attribute.



✓

EXERCISE 2: FACTORY APPLICATION— ENTITY GENERALIZATION COVERAGE

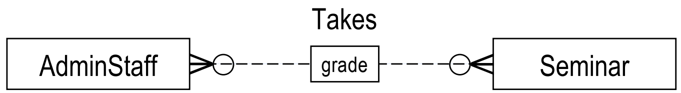
- Each employee must be an admin staff or a worker, but not both.



What should be the coverage constraint? $\Longrightarrow$ disjoint, total

EXERCISE 2: FACTORY APPLICATION—RELATIONSHIP CARDINALITY & PARTICIPATION

- From time to time, admin staff are required to take seminars.



What should be the cardinality constraints?

⟹ **Seminar** many (A seminar can be taken by many admin staff—common sense.)

AdminStaff many (An admin staff can take many seminars—common sense.)

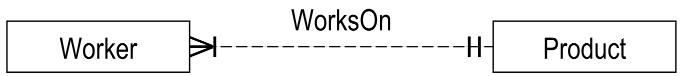
What should be the participation constraints?

⟹ **Seminar** partial (A Seminar instance should be able to exist *before* any admin staff take it.)

AdminStaff partial (Although admin staff are required to take seminars, an AdminStaff instance can exist *before* having taken any seminar.)

EXERCISE 2: FACTORY APPLICATION—RELATIONSHIP CARDINALITY & PARTICIPATION

- A worker is assigned to work on exactly one product; a product has multiple (one or more) workers assigned to it.



What should be the cardinality constraints?

⟹ **Worker** one (A worker is assigned to work on exactly one product.)
Product many (A product has multiple (one or more) workers assigned to it.)

What should be the participation constraints?

⟹ **Worker** total (A worker is assigned to work on exactly one product.)
Product total (A product has multiple (one or more) workers assigned to it.)

EXERCISE 2: FACTORY APPLICATION—RELATIONSHIP CARDINALITY & PARTICIPATION

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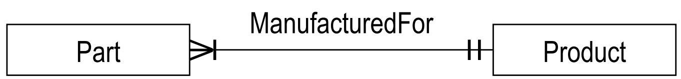

What should be the cardinality constraints?

⟹ **Part** one (A part can physically belong to only one product—common sense.)
Product many (Several parts are manufactured for each product.)

What should be the participation constraints?

⟹ **Part** total (Each part is for some product—common sense.)
Product total (A product has at least one part—common sense.)

EXERCISE 2: FACTORY APPLICATION— E-R DIAGRAM